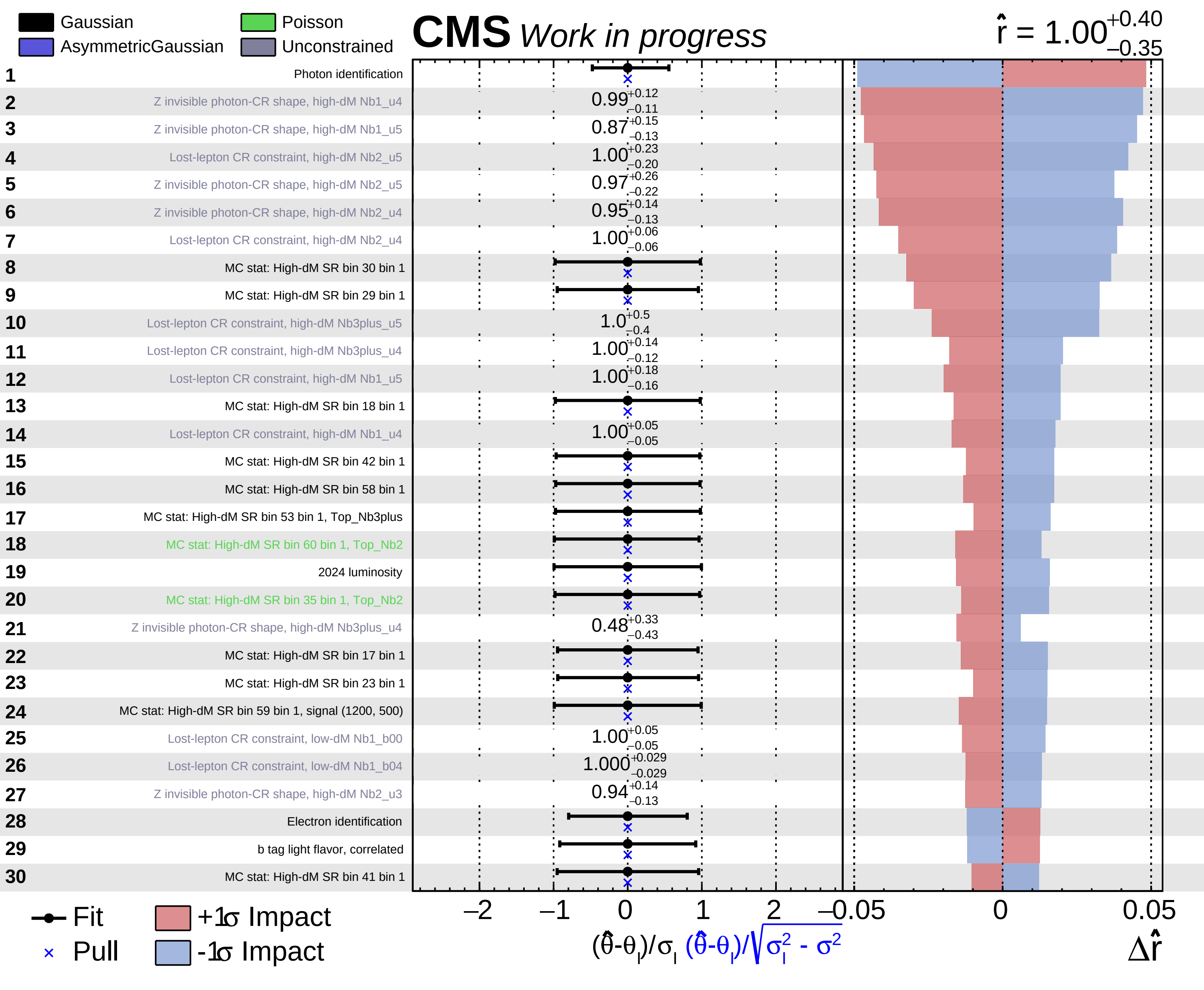
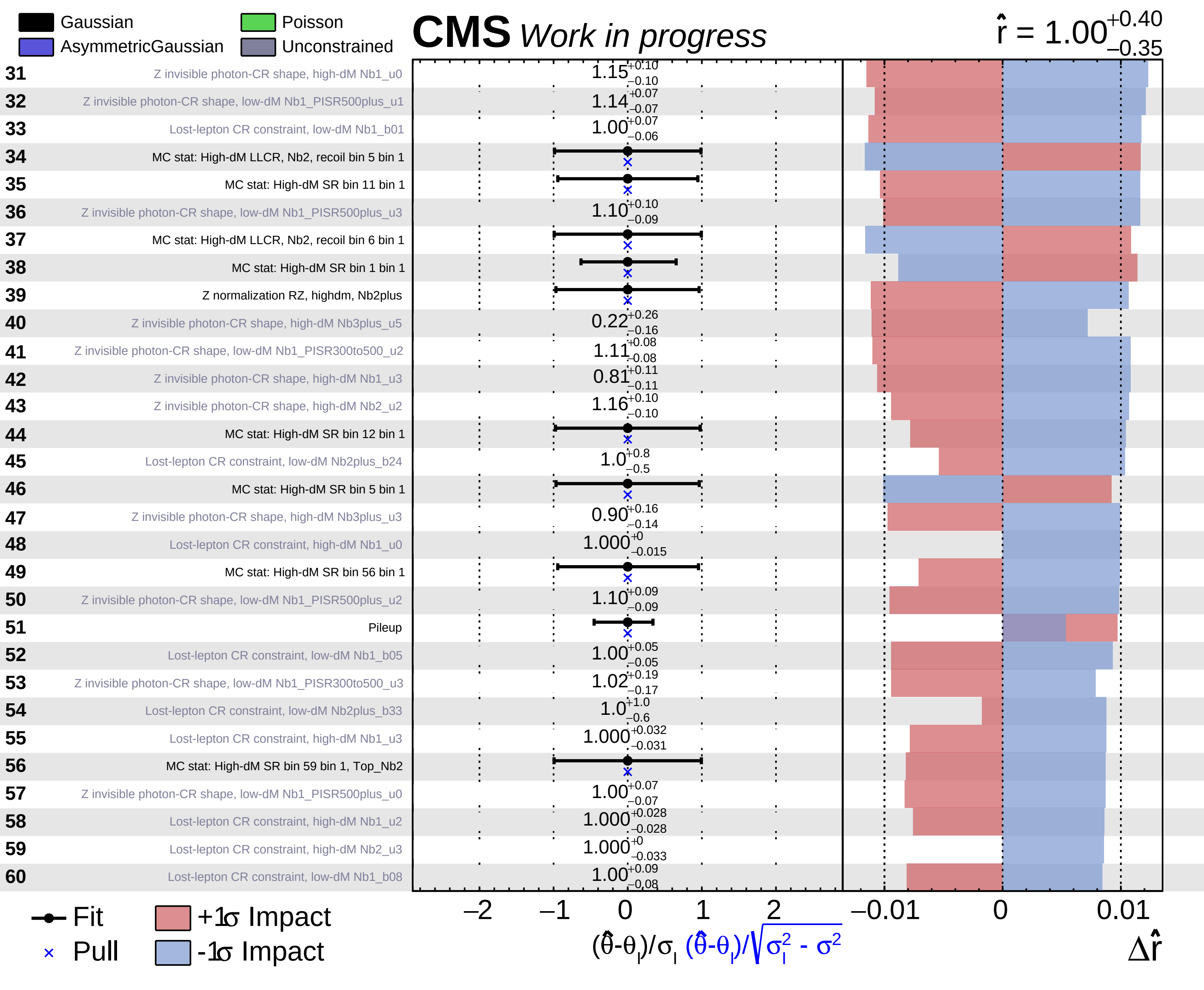
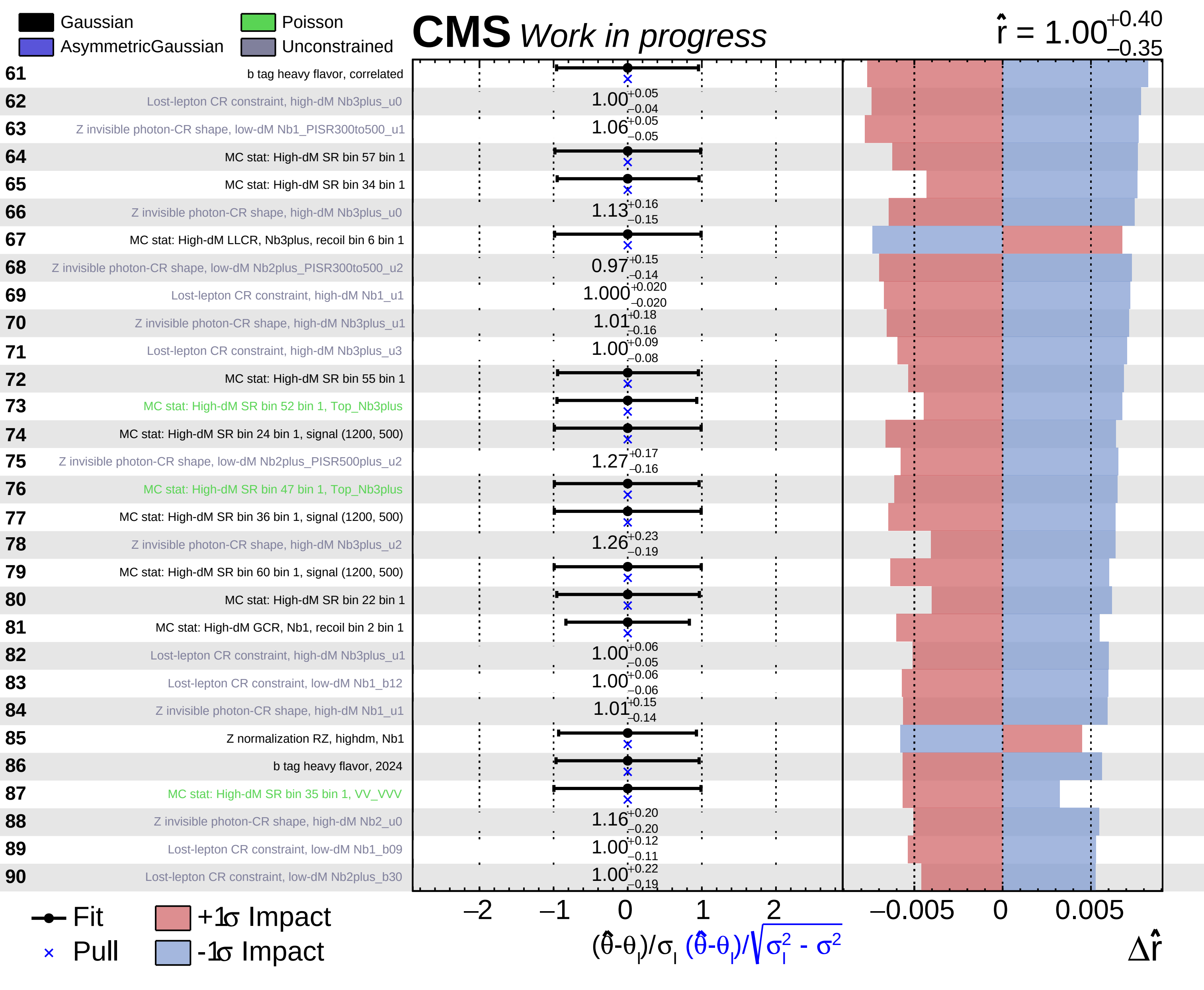
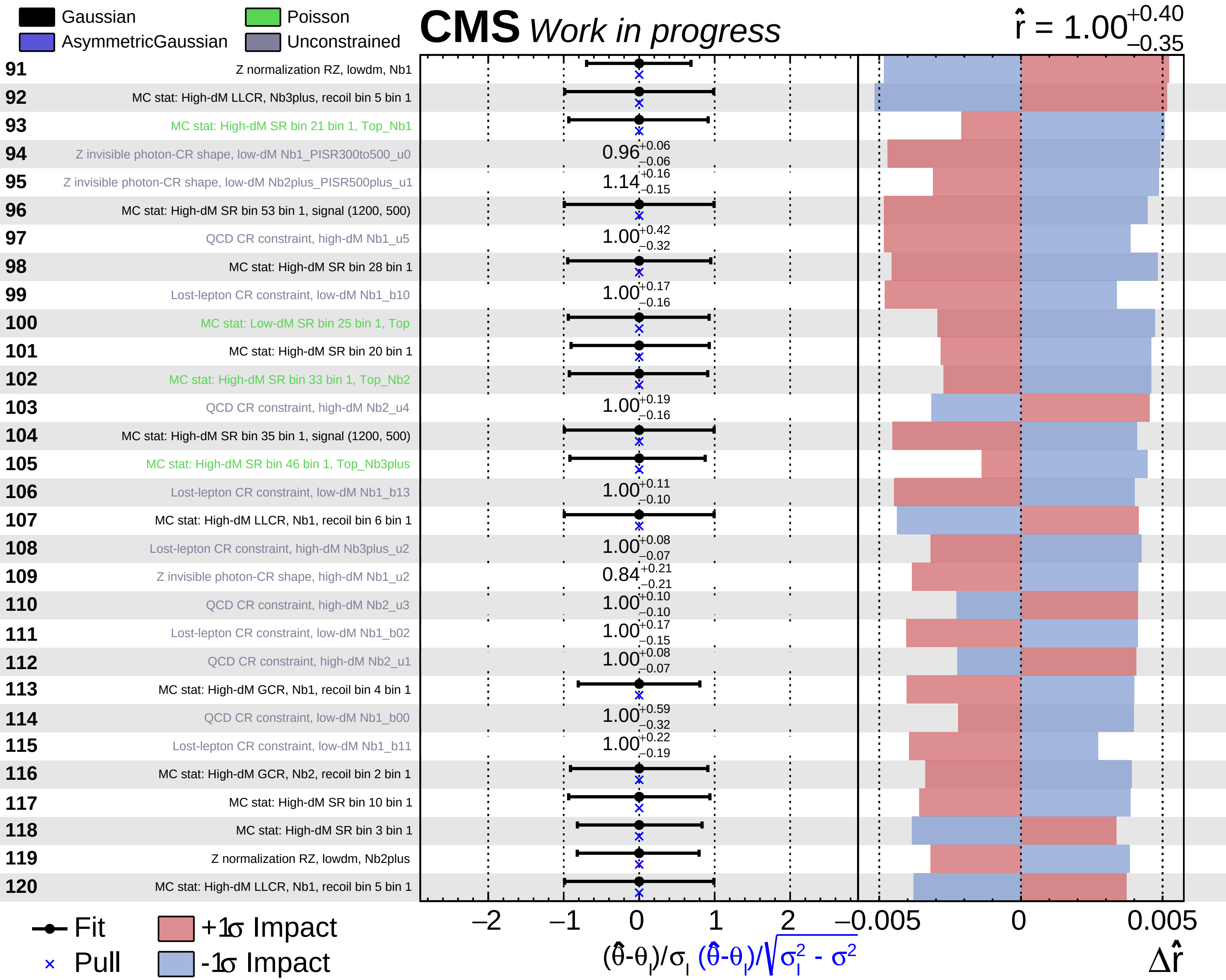


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
1 Photon identification
2 Z invisible photon-CR shape, high-dM Nb1_u4 0.99 +0.12 −0.11
3 Z invisible photon-CR shape, high-dM Nb1_u5 0.87 +0.15 −0.13
4 Lost-lepton CR constraint, high-dM Nb2_u5 1.00 +0.23 −0.20
5 Z invisible photon-CR shape, high-dM Nb2_u5 0.97 +0.26 −0.22
6 Z invisible photon-CR shape, high-dM Nb2_u4 0.95 +0.14 −0.13
7 Lost-lepton CR constraint, high-dM Nb2_u4 1.00 +0.06 −0.06
8 MC stat: High-dM SR bin 30 bin 1
9 MC stat: High-dM SR bin 29 bin 1
10 Lost-lepton CR constraint, high-dM Nb3plus_u5 1.0 +0.5 −0.4
11 Lost-lepton CR constraint, high-dM Nb3plus_u4 1.00 +0.14 −0.12
12 Lost-lepton CR constraint, high-dM Nb1_u5 1.00 +0.18 −0.16
13 MC stat: High-dM SR bin 18 bin 1
14 Lost-lepton CR constraint, high-dM Nb1_u4 1.00 +0.05 −0.05
15 MC stat: High-dM SR bin 42 bin 1
16 MC stat: High-dM SR bin 58 bin 1
17 MC stat: High-dM SR bin 53 bin 1, Top_Nb3plus
18 MC stat: High-dM SR bin 60 bin 1, Top_Nb2
19 2024 luminosity
20 MC stat: High-dM SR bin 35 bin 1, Top_Nb2
21 Z invisible photon-CR shape, high-dM Nb3plus_u4 0.48 +0.33 −0.43
22 MC stat: High-dM SR bin 17 bin 1
23 MC stat: High-dM SR bin 23 bin 1
24 MC stat: High-dM SR bin 59 bin 1, signal (1200, 500)
25 Lost-lepton CR constraint, low-dM Nb1_b00 1.00 +0.05 −0.05
26 Lost-lepton CR constraint, low-dM Nb1_b04 1.000 +0.029 −0.029
27 Z invisible photon-CR shape, high-dM Nb2_u3 0.94 +0.14 −0.13
28 Electron identification
29 b tag light flavor, correlated
30 MC stat: High-dM SR bin 41 bin 1
−2 −1 0 1 2
−0.05 0 0.05
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

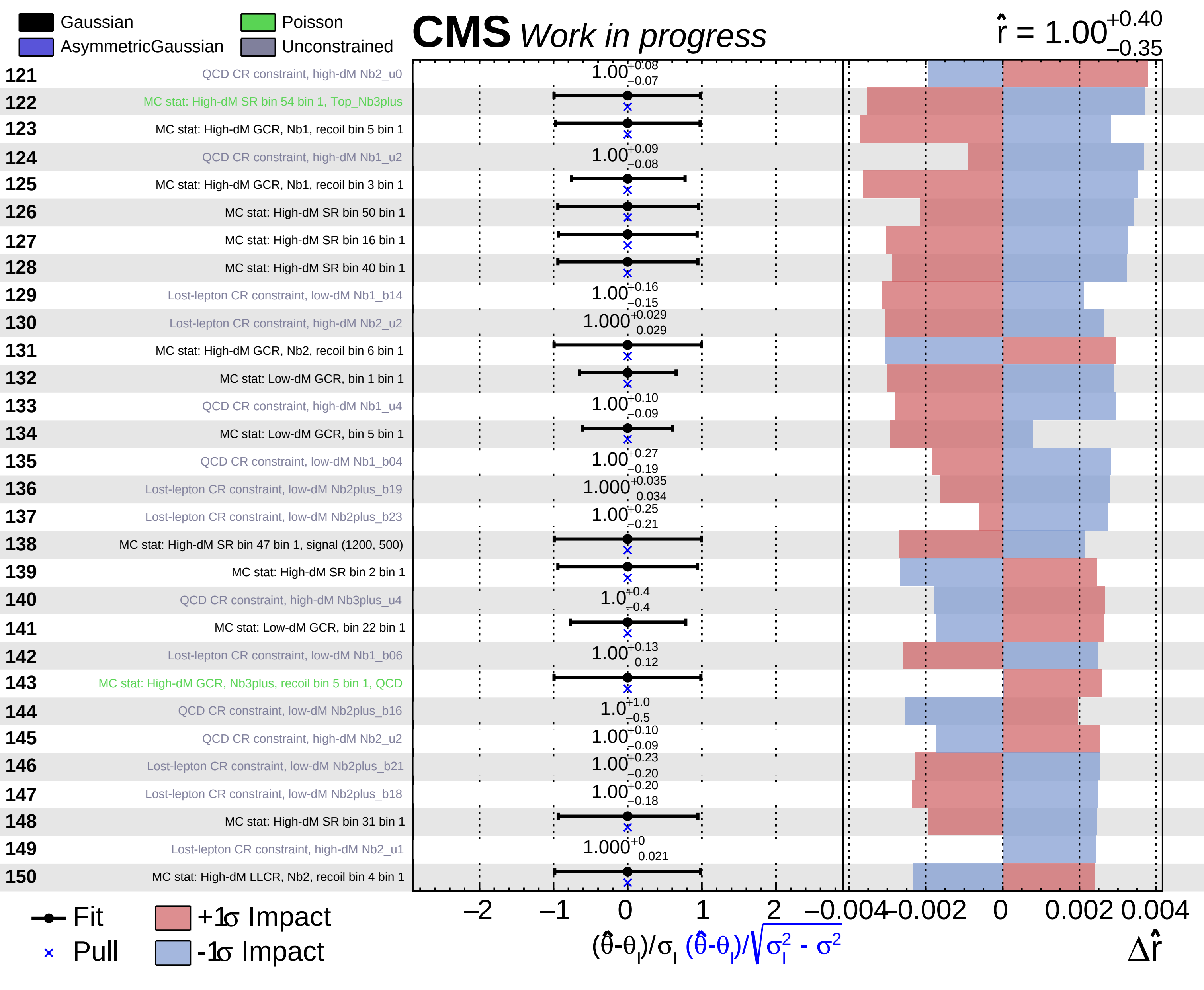

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
31 Z invisible photon-CR shape, high-dM Nb1_u0 1.15 +0.10 −0.10
32 Z invisible photon-CR shape, low-dM Nb1_PISR500plus_u1 1.14 +0.07 −0.07
33 Lost-lepton CR constraint, low-dM Nb1_b01 1.00 +0.07 −0.06
34 MC stat: High-dM LLCR, Nb2, recoil bin 5 bin 1
35 MC stat: High-dM SR bin 11 bin 1
36 Z invisible photon-CR shape, low-dM Nb1_PISR500plus_u3 1.10 +0.10 −0.09
37 MC stat: High-dM LLCR, Nb2, recoil bin 6 bin 1
38 MC stat: High-dM SR bin 1 bin 1
39 Z normalization RZ, highdm, Nb2plus
40 Z invisible photon-CR shape, high-dM Nb3plus_u5 0.22 +0.26 −0.16
41 Z invisible photon-CR shape, low-dM Nb1_PISR300to500_u2 1.11 +0.08 −0.08
42 Z invisible photon-CR shape, high-dM Nb1_u3 0.81 +0.11 −0.11
43 Z invisible photon-CR shape, high-dM Nb2_u2 1.16 +0.10 −0.10
44 MC stat: High-dM SR bin 12 bin 1
45 Lost-lepton CR constraint, low-dM Nb2plus_b24 1.0 +0.8 −0.5
46 MC stat: High-dM SR bin 5 bin 1
47 Z invisible photon-CR shape, high-dM Nb3plus_u3 0.90 +0.16 −0.14
48 Lost-lepton CR constraint, high-dM Nb1_u0 1.000 +0 −0.015
49 MC stat: High-dM SR bin 56 bin 1
50 Z invisible photon-CR shape, low-dM Nb1_PISR500plus_u2 1.10 +0.09 −0.09
51 Pileup
52 Lost-lepton CR constraint, low-dM Nb1_b05 1.00 +0.05 −0.05
53 Z invisible photon-CR shape, low-dM Nb1_PISR300to500_u3 1.02 +0.19 −0.17
54 Lost-lepton CR constraint, low-dM Nb2plus_b33 1.0 +1.0 −0.6
55 Lost-lepton CR constraint, high-dM Nb1_u3 1.000 +0.032 −0.031
56 MC stat: High-dM SR bin 59 bin 1, Top_Nb2
57 Z invisible photon-CR shape, low-dM Nb1_PISR500plus_u0 1.00 +0.07 −0.07
58 Lost-lepton CR constraint, high-dM Nb1_u2 1.000 +0.028 −0.028
59 Lost-lepton CR constraint, high-dM Nb2_u3 1.000 +0 −0.033
60 Lost-lepton CR constraint, low-dM Nb1_b08 1.00 +0.09 −0.08
−2 −1 0 1 2
−0.01 0 0.01
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

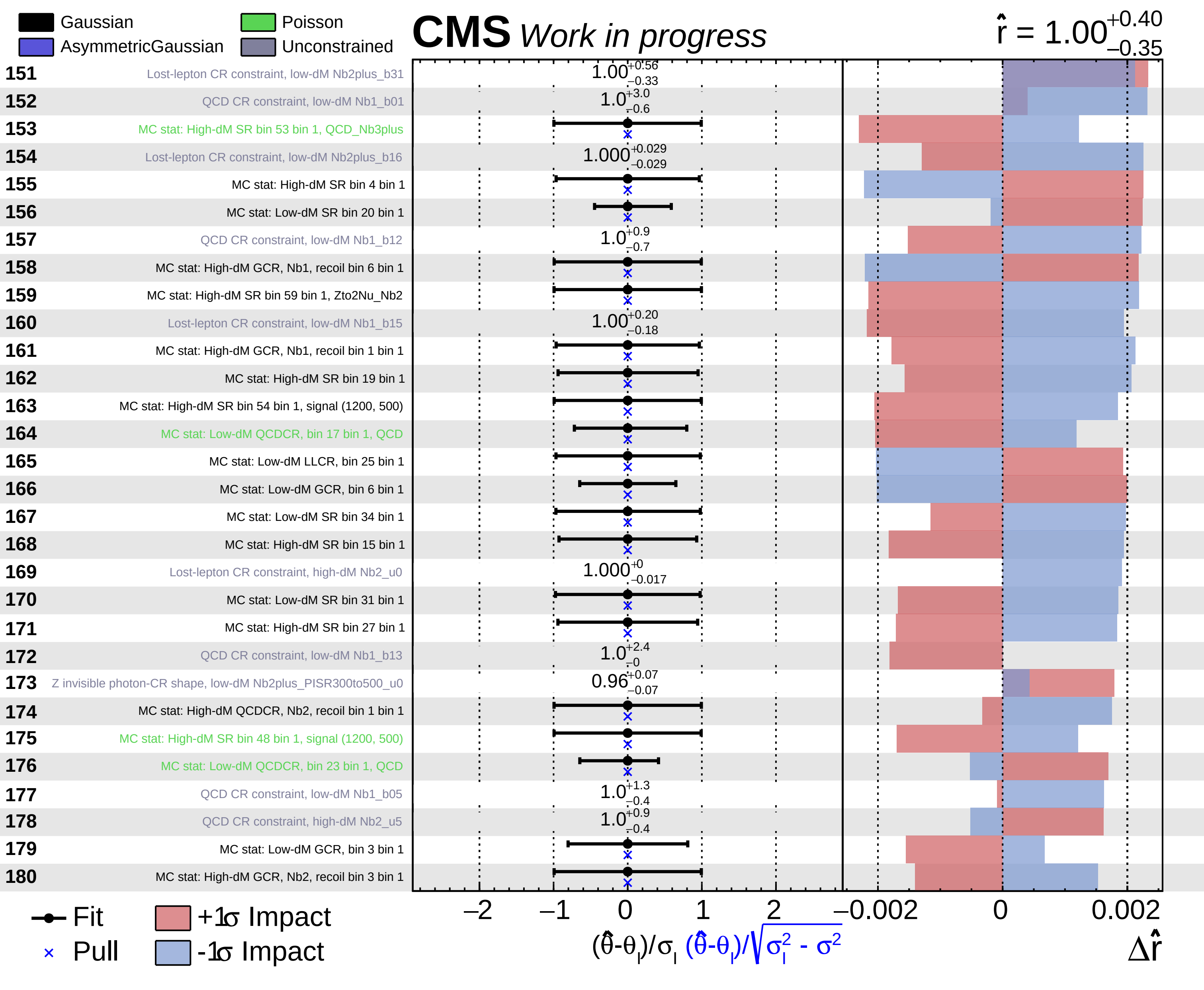


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
61 b tag heavy flavor, correlated
62 Lost-lepton CR constraint, high-dM Nb3plus_u0 $1.00^{+0.05}_{-0.04}$
63 Z invisible photon-CR shape, low-dM Nb1_PISR300to500_u1 $1.06^{+0.05}_{-0.05}$
64 MC stat: High-dM SR bin 57 bin 1
65 MC stat: High-dM SR bin 34 bin 1
66 Z invisible photon-CR shape, high-dM Nb3plus_u0 $1.13^{+0.16}_{-0.15}$
67 MC stat: High-dM LLCR, Nb3plus, recoil bin 6 bin 1
68 Z invisible photon-CR shape, low-dM Nb2plus_PISR300to500_u2 $0.97^{+0.15}_{-0.14}$
69 Lost-lepton CR constraint, high-dM Nb1_u1 $1.000^{+0.020}_{-0.020}$
70 Z invisible photon-CR shape, high-dM Nb3plus_u1 $1.01^{+0.18}_{-0.16}$
71 Lost-lepton CR constraint, high-dM Nb3plus_u3 $1.00^{+0.09}_{-0.08}$
72 MC stat: High-dM SR bin 55 bin 1
73 MC stat: High-dM SR bin 52 bin 1, Top_Nb3plus
74 MC stat: High-dM SR bin 24 bin 1, signal (1200, 500)
75 Z invisible photon-CR shape, low-dM Nb2plus_PISR500plus_u2 $1.27^{+0.17}_{-0.16}$
76 MC stat: High-dM SR bin 47 bin 1, Top_Nb3plus
77 MC stat: High-dM SR bin 36 bin 1, signal (1200, 500)
78 Z invisible photon-CR shape, high-dM Nb3plus_u2 $1.26^{+0.23}_{-0.19}$
79 MC stat: High-dM SR bin 60 bin 1, signal (1200, 500)
80 MC stat: High-dM SR bin 22 bin 1
81 MC stat: High-dM GCR, Nb1, recoil bin 2 bin 1
82 Lost-lepton CR constraint, high-dM Nb3plus_u1 $1.00^{+0.06}_{-0.05}$
83 Lost-lepton CR constraint, low-dM Nb1_b12 $1.00^{+0.06}_{-0.06}$
84 Z invisible photon-CR shape, high-dM Nb1_u1 $1.01^{+0.15}_{-0.14}$
85 Z normalization RZ, highdm, Nb1
86 b tag heavy flavor, 2024
87 MC stat: High-dM SR bin 35 bin 1, VV_VVV
88 Z invisible photon-CR shape, high-dM Nb2_u0 $1.16^{+0.20}_{-0.20}$
89 Lost-lepton CR constraint, low-dM Nb1_b09 $1.00^{+0.12}_{-0.11}$
90 Lost-lepton CR constraint, low-dM Nb2plus_b30 $1.00^{+0.22}_{-0.19}$
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

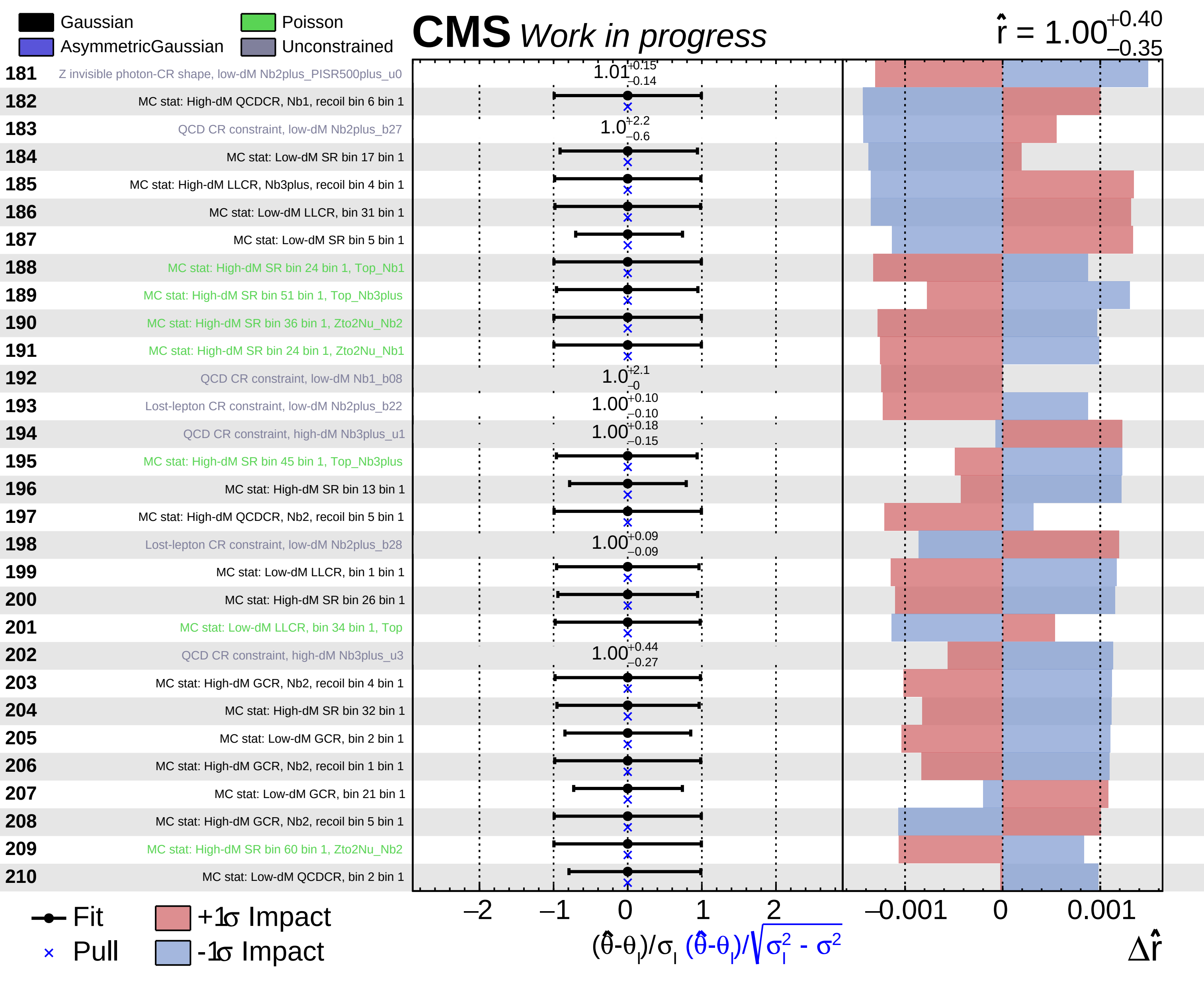


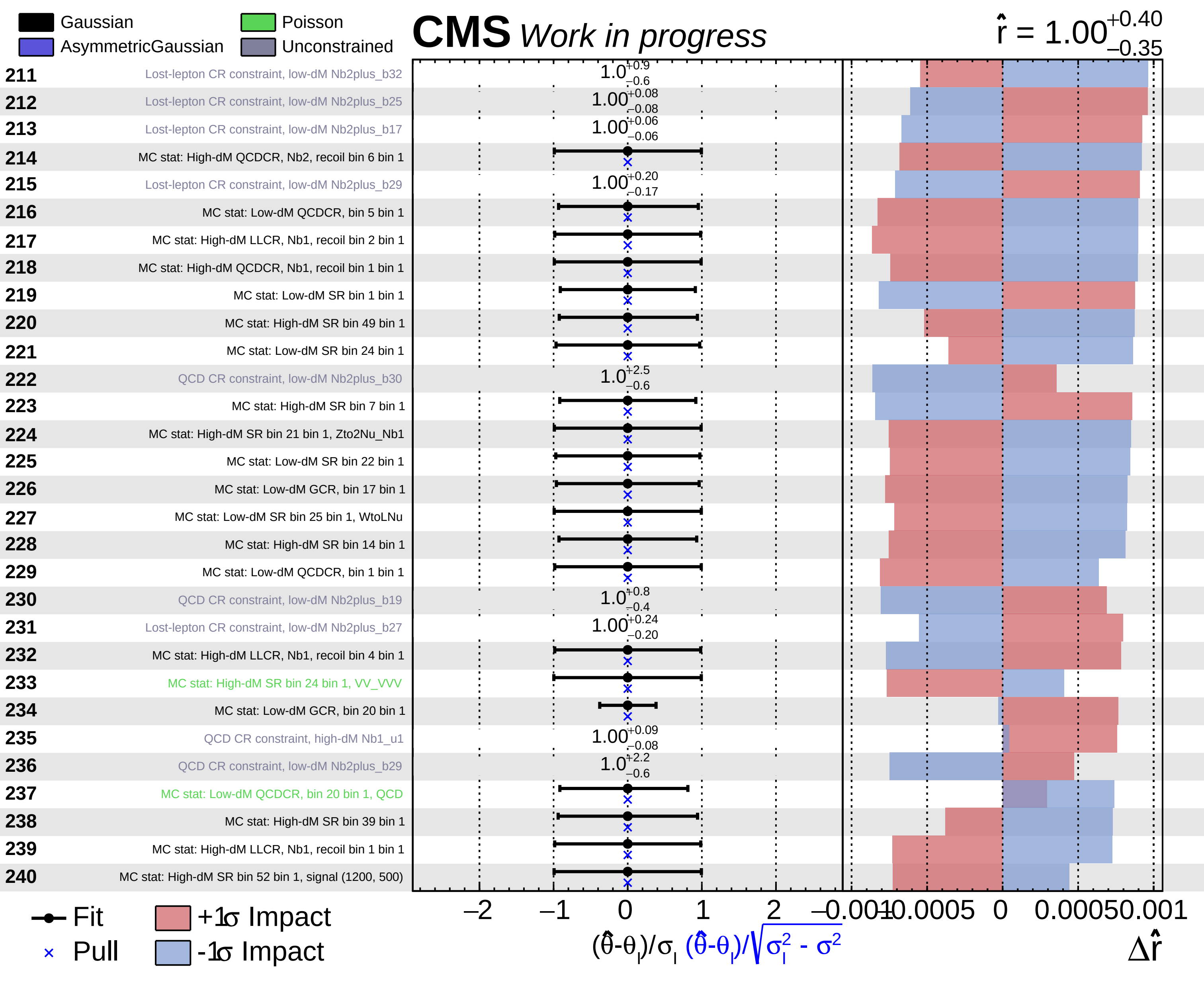
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
91 Z normalization RZ, lowdm, Nb1
92 MC stat: High-dM LLCR, Nb3plus, recoil bin 5 bin 1
93 MC stat: High-dM SR bin 21 bin 1, Top_Nb1
94 Z invisible photon-CR shape, low-dM Nb1_PISR300to500_u0 $0.96^{+0.06}_{-0.06}$
95 Z invisible photon-CR shape, low-dM Nb2plus_PISR500plus_u1 $1.14^{+0.16}_{-0.15}$
96 MC stat: High-dM SR bin 53 bin 1, signal (1200, 500)
97 QCD CR constraint, high-dM Nb1_u5 $1.00^{+0.42}_{-0.32}$
98 MC stat: High-dM SR bin 28 bin 1
99 Lost-lepton CR constraint, low-dM Nb1_b10 $1.00^{+0.17}_{-0.16}$
100 MC stat: Low-dM SR bin 25 bin 1, Top
101 MC stat: High-dM SR bin 20 bin 1
102 MC stat: High-dM SR bin 33 bin 1, Top_Nb2
103 QCD CR constraint, high-dM Nb2_u4 $1.00^{+0.19}_{-0.16}$
104 MC stat: High-dM SR bin 35 bin 1, signal (1200, 500)
105 MC stat: High-dM SR bin 46 bin 1, Top_Nb3plus
106 Lost-lepton CR constraint, low-dM Nb1_b13 $1.00^{+0.11}_{-0.10}$
107 MC stat: High-dM LLCR, Nb1, recoil bin 6 bin 1
108 Lost-lepton CR constraint, high-dM Nb3plus_u2 $1.00^{+0.08}_{-0.07}$
109 Z invisible photon-CR shape, high-dM Nb1_u2 $0.84^{+0.21}_{-0.21}$
110 QCD CR constraint, high-dM Nb2_u3 $1.00^{+0.10}_{-0.10}$
111 Lost-lepton CR constraint, low-dM Nb1_b02 $1.00^{+0.17}_{-0.15}$
112 QCD CR constraint, high-dM Nb2_u1 $1.00^{+0.08}_{-0.07}$
113 MC stat: High-dM GCR, Nb1, recoil bin 4 bin 1
114 QCD CR constraint, low-dM Nb1_b00 $1.00^{+0.59}_{-0.32}$
115 Lost-lepton CR constraint, low-dM Nb1_b11 $1.00^{+0.22}_{-0.19}$
116 MC stat: High-dM GCR, Nb2, recoil bin 2 bin 1
117 MC stat: High-dM SR bin 10 bin 1
118 MC stat: High-dM SR bin 3 bin 1
119 Z normalization RZ, lowdm, Nb2plus
120 MC stat: High-dM LLCR, Nb1, recoil bin 5 bin 1
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
121 QCD CR constraint, high-dM Nb2_u0
$1.00^{+0.08}_{-0.07}$
122 MC stat: High-dM SR bin 54 bin 1, Top_Nb3plus
123 MC stat: High-dM GCR, Nb1, recoil bin 5 bin 1
124 QCD CR constraint, high-dM Nb1_u2
$1.00^{+0.09}_{-0.08}$
125 MC stat: High-dM GCR, Nb1, recoil bin 3 bin 1
126 MC stat: High-dM SR bin 50 bin 1
127 MC stat: High-dM SR bin 16 bin 1
128 MC stat: High-dM SR bin 40 bin 1
129 Lost-lepton CR constraint, low-dM Nb1_b14
$1.00^{+0.16}_{-0.15}$
130 Lost-lepton CR constraint, high-dM Nb2_u2
$1.000^{+0.029}_{-0.029}$
131 MC stat: High-dM GCR, Nb2, recoil bin 6 bin 1
132 MC stat: Low-dM GCR, bin 1 bin 1
133 QCD CR constraint, high-dM Nb1_u4
$1.00^{+0.10}_{-0.09}$
134 MC stat: Low-dM GCR, bin 5 bin 1
135 QCD CR constraint, low-dM Nb1_b04
$1.00^{+0.27}_{-0.19}$
136 Lost-lepton CR constraint, low-dM Nb2plus_b19
$1.000^{+0.035}_{-0.034}$
137 Lost-lepton CR constraint, low-dM Nb2plus_b23
$1.00^{+0.25}_{-0.21}$
138 MC stat: High-dM SR bin 47 bin 1, signal (1200, 500)
139 MC stat: High-dM SR bin 2 bin 1
140 QCD CR constraint, high-dM Nb3plus_u4
$1.0^{+0.4}_{-0.4}$
141 MC stat: Low-dM GCR, bin 22 bin 1
142 Lost-lepton CR constraint, low-dM Nb1_b06
$1.00^{+0.13}_{-0.12}$
143 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, QCD
144 QCD CR constraint, low-dM Nb2plus_b16
$1.0^{+1.0}_{-0.5}$
145 QCD CR constraint, high-dM Nb2_u2
$1.00^{+0.10}_{-0.09}$
146 Lost-lepton CR constraint, low-dM Nb2plus_b21
$1.00^{+0.23}_{-0.20}$
147 Lost-lepton CR constraint, low-dM Nb2plus_b18
$1.00^{+0.20}_{-0.18}$
148 MC stat: High-dM SR bin 31 bin 1
149 Lost-lepton CR constraint, high-dM Nb2_u1
$1.000^{+0}_{-0.021}$
150 MC stat: High-dM LLCR, Nb2, recoil bin 4 bin 1
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

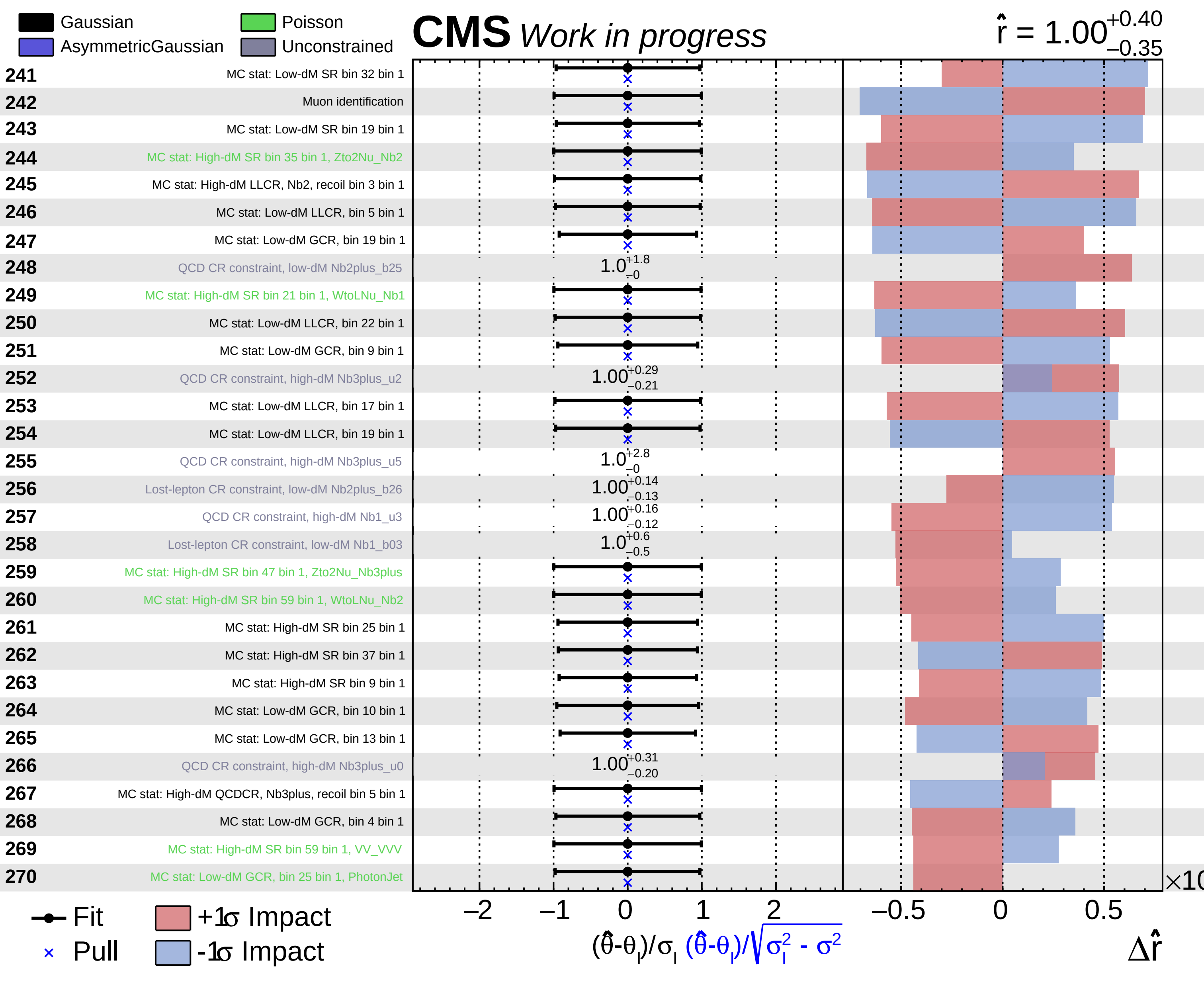


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
151 Lost-lepton CR constraint, low-dM Nb2plus_b31
152 QCD CR constraint, low-dM Nb1_b01
153 MC stat: High-dM SR bin 53 bin 1, QCD_Nb3plus
154 Lost-lepton CR constraint, low-dM Nb2plus_b16
155 MC stat: High-dM SR bin 4 bin 1
156 MC stat: Low-dM SR bin 20 bin 1
157 QCD CR constraint, low-dM Nb1_b12
158 MC stat: High-dM GCR, Nb1, recoil bin 6 bin 1
159 MC stat: High-dM SR bin 59 bin 1, Zto2Nu_Nb2
160 Lost-lepton CR constraint, low-dM Nb1_b15
161 MC stat: High-dM GCR, Nb1, recoil bin 1 bin 1
162 MC stat: High-dM SR bin 19 bin 1
163 MC stat: High-dM SR bin 54 bin 1, signal (1200, 500)
164 MC stat: Low-dM QCDCR, bin 17 bin 1, QCD
165 MC stat: Low-dM LLCR, bin 25 bin 1
166 MC stat: Low-dM GCR, bin 6 bin 1
167 MC stat: Low-dM SR bin 34 bin 1
168 MC stat: High-dM SR bin 15 bin 1
169 Lost-lepton CR constraint, high-dM Nb2_u0
170 MC stat: Low-dM SR bin 31 bin 1
171 MC stat: High-dM SR bin 27 bin 1
172 QCD CR constraint, low-dM Nb1_b13
173 Z invisible photon-CR shape, low-dM Nb2plus_PISR300to500_u0
174 MC stat: High-dM QCDCR, Nb2, recoil bin 1 bin 1
175 MC stat: High-dM SR bin 48 bin 1, signal (1200, 500)
176 MC stat: Low-dM QCDCR, bin 23 bin 1, QCD
177 QCD CR constraint, low-dM Nb1_b05
178 QCD CR constraint, high-dM Nb2_u5
179 MC stat: Low-dM GCR, bin 3 bin 1
180 MC stat: High-dM GCR, Nb2, recoil bin 3 bin 1
1.00 +0.56 −0.33
1.0 +3.0 −0.6
1.000 +0.029 −0.029
1.0 +0.9 −0.7
1.00 +0.20 −0.18
1.000 +0 −0.017
1.0 +2.4 −0
0.96 +0.07 −0.07
1.0 +1.3 −0.4
1.0 +0.9 −0.4
−2 −1 0 1 2
−0.002 0 0.002
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

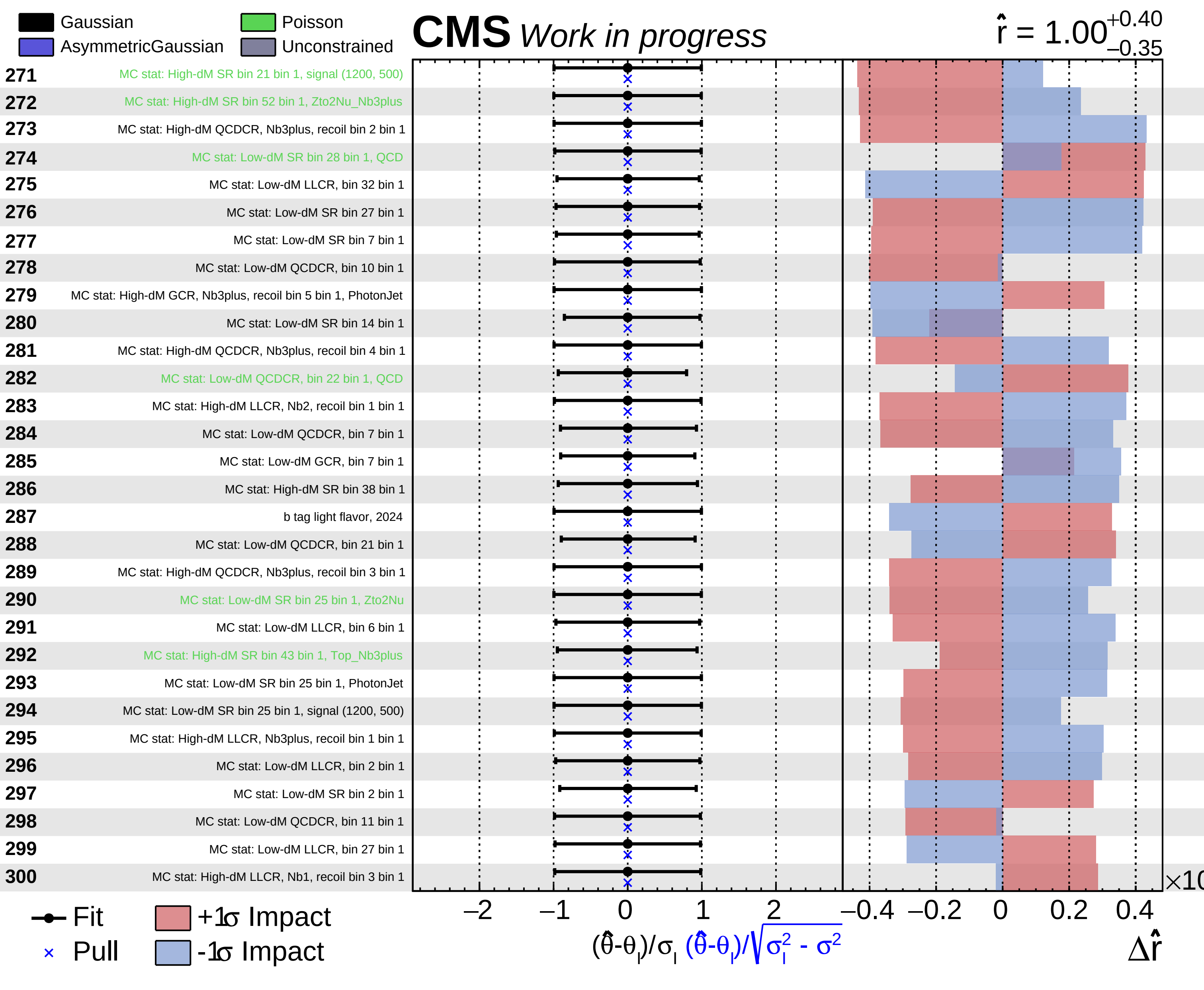

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
181 Z invisible photon-CR shape, low-dM Nb2plus_PISR500plus_u0 1.01 +0.15 −0.14
182 MC stat: High-dM QCDCR, Nb1, recoil bin 6 bin 1
183 QCD CR constraint, low-dM Nb2plus_b27 1.0 +2.2 −0.6
184 MC stat: Low-dM SR bin 17 bin 1
185 MC stat: High-dM LLCR, Nb3plus, recoil bin 4 bin 1
186 MC stat: Low-dM LLCR, bin 31 bin 1
187 MC stat: Low-dM SR bin 5 bin 1
188 MC stat: High-dM SR bin 24 bin 1, Top_Nb1
189 MC stat: High-dM SR bin 51 bin 1, Top_Nb3plus
190 MC stat: High-dM SR bin 36 bin 1, Zto2Nu_Nb2
191 MC stat: High-dM SR bin 24 bin 1, Zto2Nu_Nb1
192 QCD CR constraint, low-dM Nb1_b08 1.0 +2.1 −0
193 Lost-lepton CR constraint, low-dM Nb2plus_b22 1.00 +0.10 −0.10
194 QCD CR constraint, high-dM Nb3plus_u1 1.00 +0.18 −0.15
195 MC stat: High-dM SR bin 45 bin 1, Top_Nb3plus
196 MC stat: High-dM SR bin 13 bin 1
197 MC stat: High-dM QCDCR, Nb2, recoil bin 5 bin 1
198 Lost-lepton CR constraint, low-dM Nb2plus_b28 1.00 +0.09 −0.09
199 MC stat: Low-dM LLCR, bin 1 bin 1
200 MC stat: High-dM SR bin 26 bin 1
201 MC stat: Low-dM LLCR, bin 34 bin 1, Top
202 QCD CR constraint, high-dM Nb3plus_u3 1.00 +0.44 −0.27
203 MC stat: High-dM GCR, Nb2, recoil bin 4 bin 1
204 MC stat: High-dM SR bin 32 bin 1
205 MC stat: Low-dM GCR, bin 2 bin 1
206 MC stat: High-dM GCR, Nb2, recoil bin 1 bin 1
207 MC stat: Low-dM GCR, bin 21 bin 1
208 MC stat: High-dM GCR, Nb2, recoil bin 5 bin 1
209 MC stat: High-dM SR bin 60 bin 1, Zto2Nu_Nb2
210 MC stat: Low-dM QCDCR, bin 2 bin 1
−2 −1 0 1 2
−0.001 0 0.001
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂



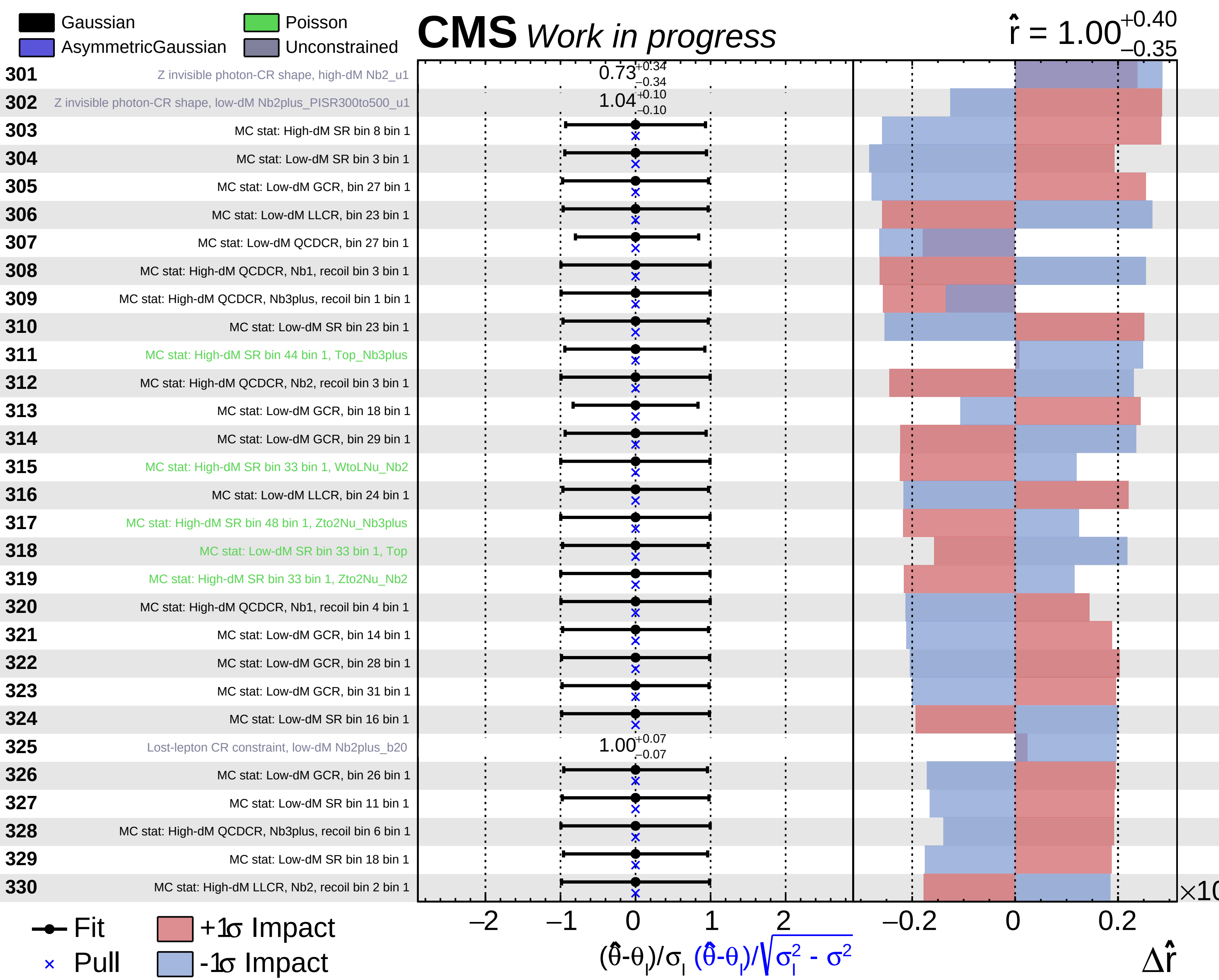
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
211 Lost-lepton CR constraint, low-dM Nb2plus_b32 $1.0^{+0.9}_{-0.6}$
212 Lost-lepton CR constraint, low-dM Nb2plus_b25 $1.00^{+0.08}_{-0.08}$
213 Lost-lepton CR constraint, low-dM Nb2plus_b17 $1.00^{+0.06}_{-0.06}$
214 MC stat: High-dM QCDCR, Nb2, recoil bin 6 bin 1
215 Lost-lepton CR constraint, low-dM Nb2plus_b29 $1.00^{+0.20}_{-0.17}$
216 MC stat: Low-dM QCDCR, bin 5 bin 1
217 MC stat: High-dM LLCR, Nb1, recoil bin 2 bin 1
218 MC stat: High-dM QCDCR, Nb1, recoil bin 1 bin 1
219 MC stat: Low-dM SR bin 1 bin 1
220 MC stat: High-dM SR bin 49 bin 1
221 MC stat: Low-dM SR bin 24 bin 1
222 QCD CR constraint, low-dM Nb2plus_b30 $1.0^{+2.5}_{-0.6}$
223 MC stat: High-dM SR bin 7 bin 1
224 MC stat: High-dM SR bin 21 bin 1, Zto2Nu_Nb1
225 MC stat: Low-dM SR bin 22 bin 1
226 MC stat: Low-dM GCR, bin 17 bin 1
227 MC stat: Low-dM SR bin 25 bin 1, WtoLNu
228 MC stat: High-dM SR bin 14 bin 1
229 MC stat: Low-dM QCDCR, bin 1 bin 1
230 QCD CR constraint, low-dM Nb2plus_b19 $1.0^{+0.8}_{-0.4}$
231 Lost-lepton CR constraint, low-dM Nb2plus_b27 $1.00^{+0.24}_{-0.20}$
232 MC stat: High-dM LLCR, Nb1, recoil bin 4 bin 1
233 MC stat: High-dM SR bin 24 bin 1, VV_VVV
234 MC stat: Low-dM GCR, bin 20 bin 1
235 QCD CR constraint, high-dM Nb1_u1 $1.00^{+0.09}_{-0.08}$
236 QCD CR constraint, low-dM Nb2plus_b29 $1.0^{+2.2}_{-0.6}$
237 MC stat: Low-dM QCDCR, bin 20 bin 1, QCD
238 MC stat: High-dM SR bin 39 bin 1
239 MC stat: High-dM LLCR, Nb1, recoil bin 1 bin 1
240 MC stat: High-dM SR bin 52 bin 1, signal (1200, 500)
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



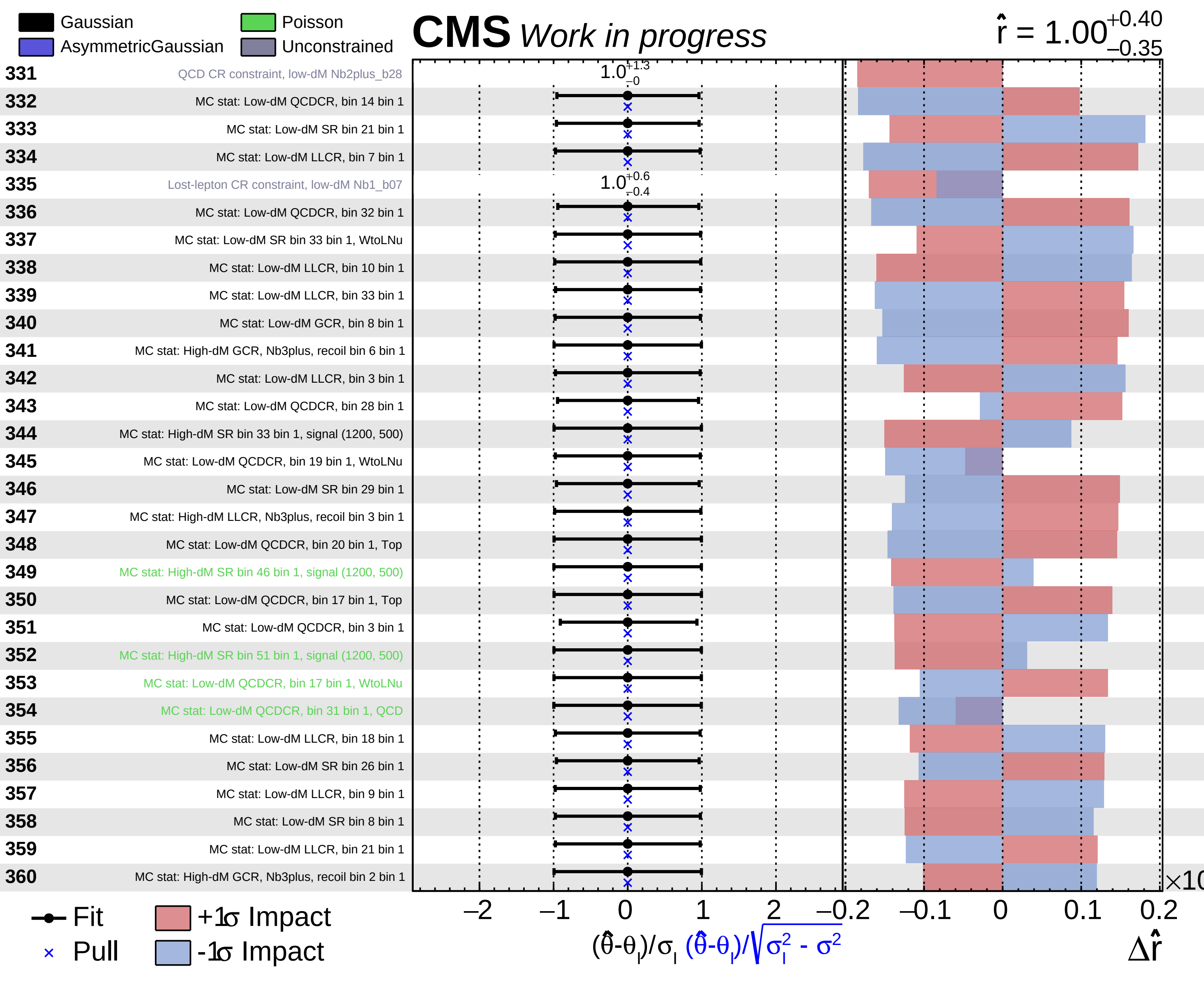
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
241 MC stat: Low-dM SR bin 32 bin 1
242 Muon identification
243 MC stat: Low-dM SR bin 19 bin 1
244 MC stat: High-dM SR bin 35 bin 1, Zto2Nu_Nb2
245 MC stat: High-dM LLCR, Nb2, recoil bin 3 bin 1
246 MC stat: Low-dM LLCR, bin 5 bin 1
247 MC stat: Low-dM GCR, bin 19 bin 1
248 QCD CR constraint, low-dM Nb2plus_b25
1.0 +1.8 −0
249 MC stat: High-dM SR bin 21 bin 1, WtoLNu_Nb1
250 MC stat: Low-dM LLCR, bin 22 bin 1
251 MC stat: Low-dM GCR, bin 9 bin 1
252 QCD CR constraint, high-dM Nb3plus_u2
1.00 +0.29 −0.21
253 MC stat: Low-dM LLCR, bin 17 bin 1
254 MC stat: Low-dM LLCR, bin 19 bin 1
255 QCD CR constraint, high-dM Nb3plus_u5
1.0 +2.8 −0
256 Lost-lepton CR constraint, low-dM Nb2plus_b26
1.00 +0.14 −0.13
257 QCD CR constraint, high-dM Nb1_u3
1.00 +0.16 −0.12
258 Lost-lepton CR constraint, low-dM Nb1_b03
1.0 +0.6 −0.5
259 MC stat: High-dM SR bin 47 bin 1, Zto2Nu_Nb3plus
260 MC stat: High-dM SR bin 59 bin 1, WtoLNu_Nb2
261 MC stat: High-dM SR bin 25 bin 1
262 MC stat: High-dM SR bin 37 bin 1
263 MC stat: High-dM SR bin 9 bin 1
264 MC stat: Low-dM GCR, bin 10 bin 1
265 MC stat: Low-dM GCR, bin 13 bin 1
266 QCD CR constraint, high-dM Nb3plus_u0
1.00 +0.31 −0.20
267 MC stat: High-dM QCDCR, Nb3plus, recoil bin 5 bin 1
268 MC stat: Low-dM GCR, bin 4 bin 1
269 MC stat: High-dM SR bin 59 bin 1, VV_VVV
270 MC stat: Low-dM GCR, bin 25 bin 1, PhotonJet
−2
−1
0
1
2
−0.5
0
0.5
×10
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



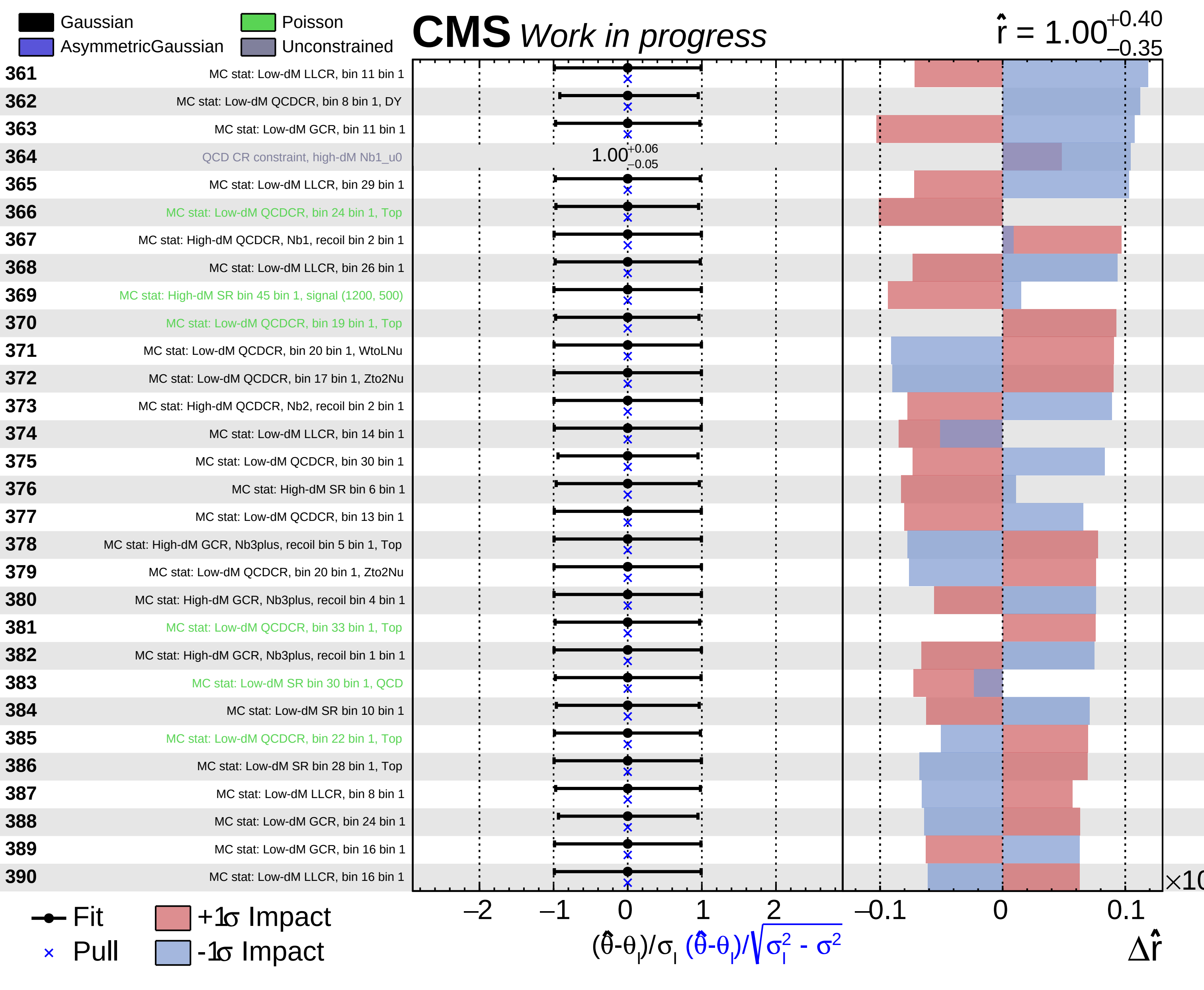
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
271 MC stat: High-dM SR bin 21 bin 1, signal (1200, 500)
272 MC stat: High-dM SR bin 52 bin 1, Zto2Nu_Nb3plus
273 MC stat: High-dM QCDCR, Nb3plus, recoil bin 2 bin 1
274 MC stat: Low-dM SR bin 28 bin 1, QCD
275 MC stat: Low-dM LLCR, bin 32 bin 1
276 MC stat: Low-dM SR bin 27 bin 1
277 MC stat: Low-dM SR bin 7 bin 1
278 MC stat: Low-dM QCDCR, bin 10 bin 1
279 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, PhotonJet
280 MC stat: Low-dM SR bin 14 bin 1
281 MC stat: High-dM QCDCR, Nb3plus, recoil bin 4 bin 1
282 MC stat: Low-dM QCDCR, bin 22 bin 1, QCD
283 MC stat: High-dM LLCR, Nb2, recoil bin 1 bin 1
284 MC stat: Low-dM QCDCR, bin 7 bin 1
285 MC stat: Low-dM GCR, bin 7 bin 1
286 MC stat: High-dM SR bin 38 bin 1
287 b tag light flavor, 2024
288 MC stat: Low-dM QCDCR, bin 21 bin 1
289 MC stat: High-dM QCDCR, Nb3plus, recoil bin 3 bin 1
290 MC stat: Low-dM SR bin 25 bin 1, Zto2Nu
291 MC stat: Low-dM LLCR, bin 6 bin 1
292 MC stat: High-dM SR bin 43 bin 1, Top_Nb3plus
293 MC stat: Low-dM SR bin 25 bin 1, PhotonJet
294 MC stat: Low-dM SR bin 25 bin 1, signal (1200, 500)
295 MC stat: High-dM LLCR, Nb3plus, recoil bin 1 bin 1
296 MC stat: Low-dM LLCR, bin 2 bin 1
297 MC stat: Low-dM SR bin 2 bin 1
298 MC stat: Low-dM QCDCR, bin 11 bin 1
299 MC stat: Low-dM LLCR, bin 27 bin 1
300 MC stat: High-dM LLCR, Nb1, recoil bin 3 bin 1
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂



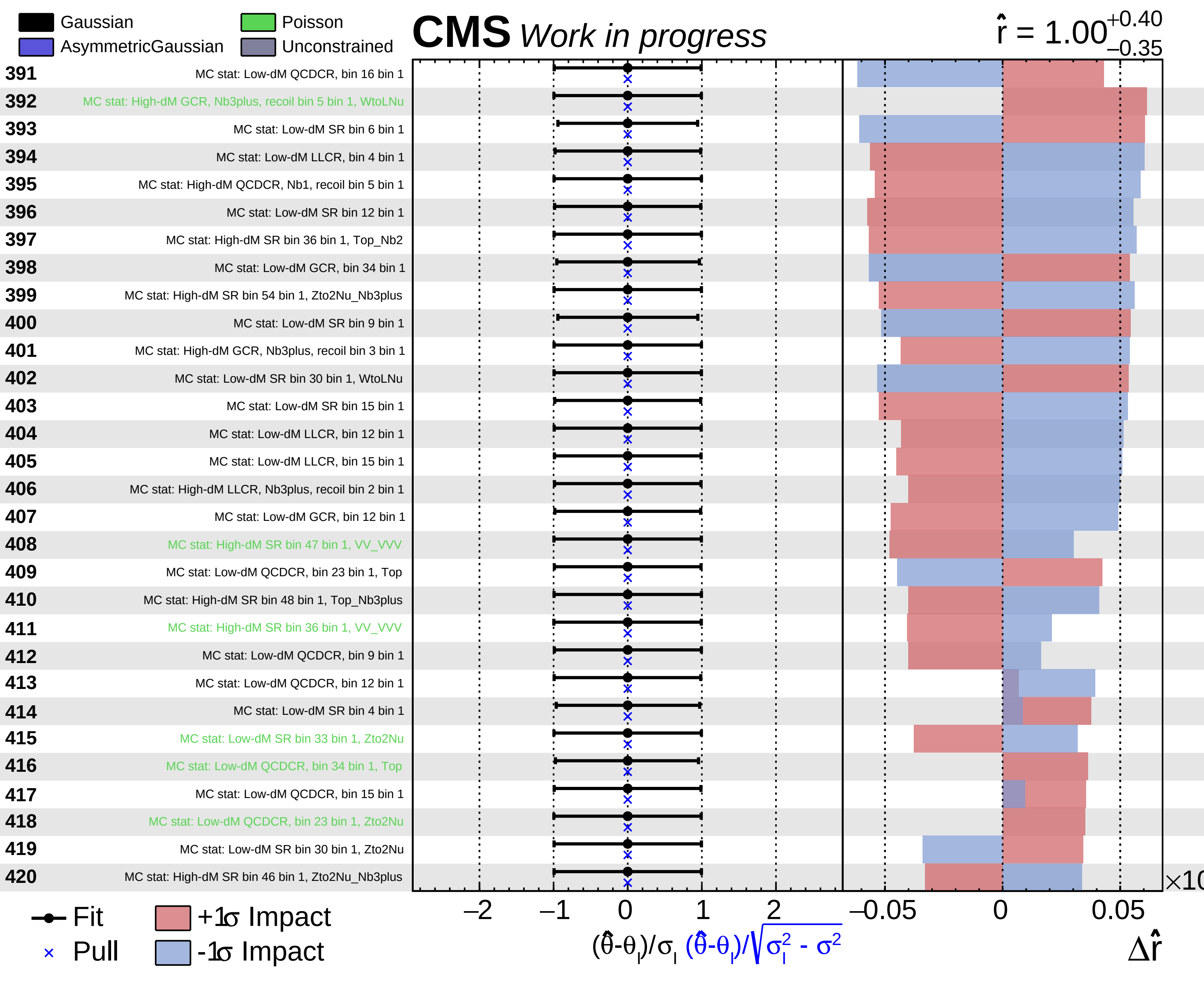
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
301 Z invisible photon-CR shape, high-dM Nb2_u1
$0.73^{+0.34}_{-0.34}$
302 Z invisible photon-CR shape, low-dM Nb2plus_PISR300to500_u1
$1.04^{+0.10}_{-0.10}$
303 MC stat: High-dM SR bin 8 bin 1
304 MC stat: Low-dM SR bin 3 bin 1
305 MC stat: Low-dM GCR, bin 27 bin 1
306 MC stat: Low-dM LLCR, bin 23 bin 1
307 MC stat: Low-dM QCDCR, bin 27 bin 1
308 MC stat: High-dM QCDCR, Nb1, recoil bin 3 bin 1
309 MC stat: High-dM QCDCR, Nb3plus, recoil bin 1 bin 1
310 MC stat: Low-dM SR bin 23 bin 1
311 MC stat: High-dM SR bin 44 bin 1, Top_Nb3plus
312 MC stat: High-dM QCDCR, Nb2, recoil bin 3 bin 1
313 MC stat: Low-dM GCR, bin 18 bin 1
314 MC stat: Low-dM GCR, bin 29 bin 1
315 MC stat: High-dM SR bin 33 bin 1, WtoLNu_Nb2
316 MC stat: Low-dM LLCR, bin 24 bin 1
317 MC stat: High-dM SR bin 48 bin 1, Zto2Nu_Nb3plus
318 MC stat: Low-dM SR bin 33 bin 1, Top
319 MC stat: High-dM SR bin 33 bin 1, Zto2Nu_Nb2
320 MC stat: High-dM QCDCR, Nb1, recoil bin 4 bin 1
321 MC stat: Low-dM GCR, bin 14 bin 1
322 MC stat: Low-dM GCR, bin 28 bin 1
323 MC stat: Low-dM GCR, bin 31 bin 1
324 MC stat: Low-dM SR bin 16 bin 1
325 Lost-lepton CR constraint, low-dM Nb2plus_b20
$1.00^{+0.07}_{-0.07}$
326 MC stat: Low-dM GCR, bin 26 bin 1
327 MC stat: Low-dM SR bin 11 bin 1
328 MC stat: High-dM QCDCR, Nb3plus, recoil bin 6 bin 1
329 MC stat: Low-dM SR bin 18 bin 1
330 MC stat: High-dM LLCR, Nb2, recoil bin 2 bin 1
−2 −1 0 1 2
−0.2 0 0.2
×10
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



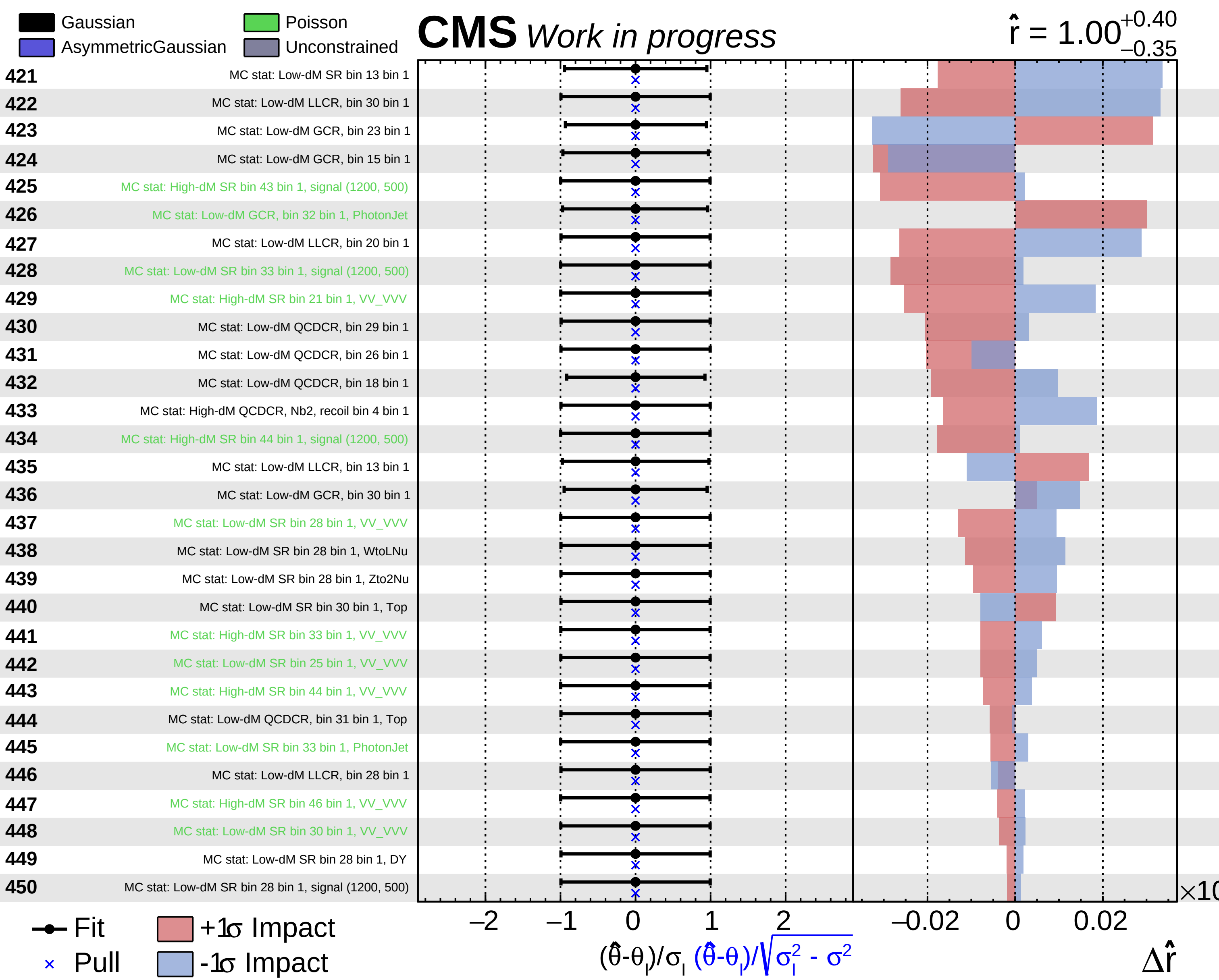
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
331 QCD CR constraint, low-dM Nb2plus_b28
$1.0^{+1.3}_{-0}$
332 MC stat: Low-dM QCDCR, bin 14 bin 1
333 MC stat: Low-dM SR bin 21 bin 1
334 MC stat: Low-dM LLCR, bin 7 bin 1
335 Lost-lepton CR constraint, low-dM Nb1_b07
$1.0^{+0.6}_{-0.4}$
336 MC stat: Low-dM QCDCR, bin 32 bin 1
337 MC stat: Low-dM SR bin 33 bin 1, WtoLNu
338 MC stat: Low-dM LLCR, bin 10 bin 1
339 MC stat: Low-dM LLCR, bin 33 bin 1
340 MC stat: Low-dM GCR, bin 8 bin 1
341 MC stat: High-dM GCR, Nb3plus, recoil bin 6 bin 1
342 MC stat: Low-dM LLCR, bin 3 bin 1
343 MC stat: Low-dM QCDCR, bin 28 bin 1
344 MC stat: High-dM SR bin 33 bin 1, signal (1200, 500)
345 MC stat: Low-dM QCDCR, bin 19 bin 1, WtoLNu
346 MC stat: Low-dM SR bin 29 bin 1
347 MC stat: High-dM LLCR, Nb3plus, recoil bin 3 bin 1
348 MC stat: Low-dM QCDCR, bin 20 bin 1, Top
349 MC stat: High-dM SR bin 46 bin 1, signal (1200, 500)
350 MC stat: Low-dM QCDCR, bin 17 bin 1, Top
351 MC stat: Low-dM QCDCR, bin 3 bin 1
352 MC stat: High-dM SR bin 51 bin 1, signal (1200, 500)
353 MC stat: Low-dM QCDCR, bin 17 bin 1, WtoLNu
354 MC stat: Low-dM QCDCR, bin 31 bin 1, QCD
355 MC stat: Low-dM LLCR, bin 18 bin 1
356 MC stat: Low-dM SR bin 26 bin 1
357 MC stat: Low-dM LLCR, bin 9 bin 1
358 MC stat: Low-dM SR bin 8 bin 1
359 MC stat: Low-dM LLCR, bin 21 bin 1
360 MC stat: High-dM GCR, Nb3plus, recoil bin 2 bin 1
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



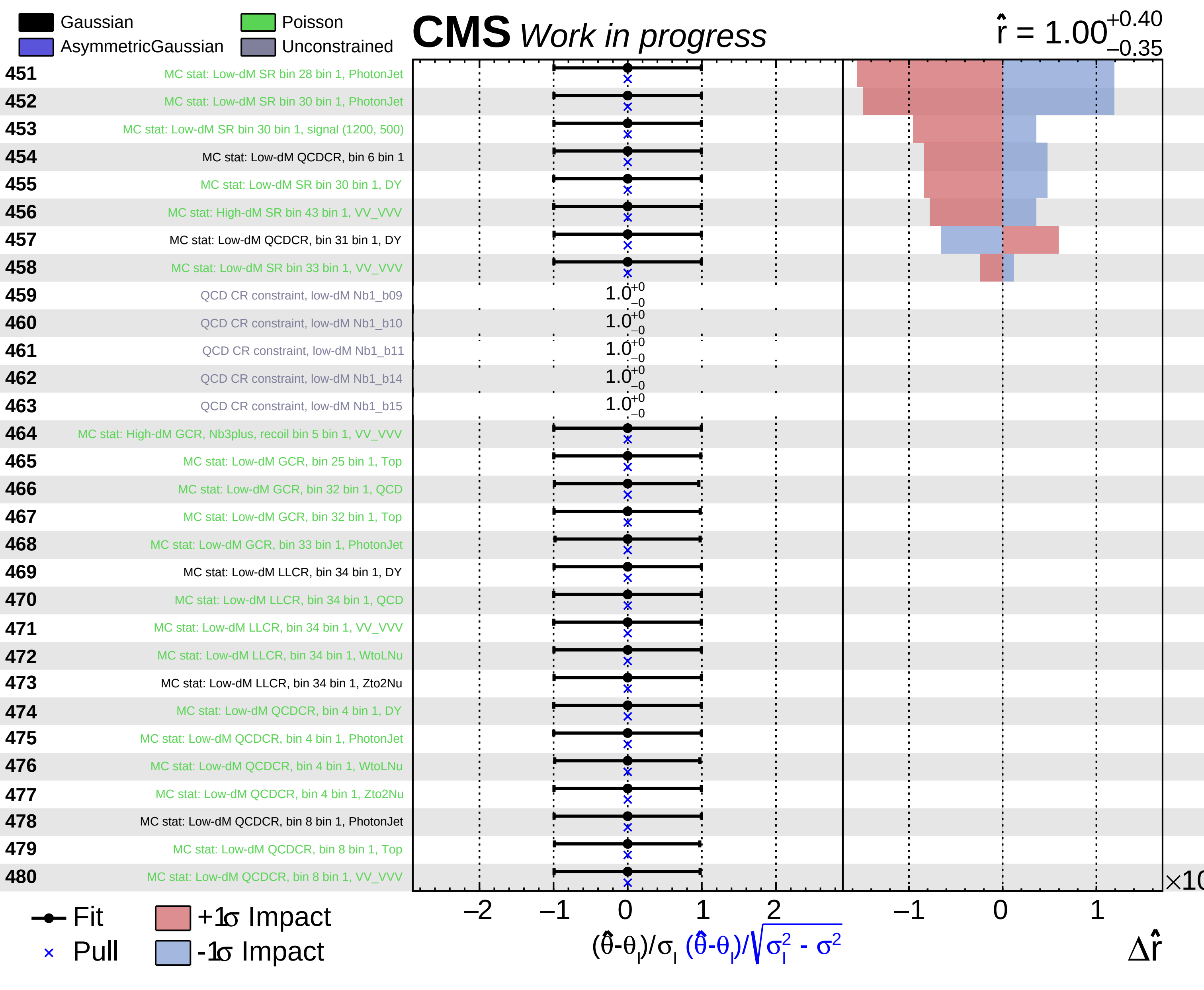
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
361 MC stat: Low-dM LLCR, bin 11 bin 1
362 MC stat: Low-dM QCDCR, bin 8 bin 1, DY
363 MC stat: Low-dM GCR, bin 11 bin 1
364 QCD CR constraint, high-dM Nb1_u0
$1.00^{+0.06}_{-0.05}$
365 MC stat: Low-dM LLCR, bin 29 bin 1
366 MC stat: Low-dM QCDCR, bin 24 bin 1, Top
367 MC stat: High-dM QCDCR, Nb1, recoil bin 2 bin 1
368 MC stat: Low-dM LLCR, bin 26 bin 1
369 MC stat: High-dM SR bin 45 bin 1, signal (1200, 500)
370 MC stat: Low-dM QCDCR, bin 19 bin 1, Top
371 MC stat: Low-dM QCDCR, bin 20 bin 1, WtoLNu
372 MC stat: Low-dM QCDCR, bin 17 bin 1, Zto2Nu
373 MC stat: High-dM QCDCR, Nb2, recoil bin 2 bin 1
374 MC stat: Low-dM LLCR, bin 14 bin 1
375 MC stat: Low-dM QCDCR, bin 30 bin 1
376 MC stat: High-dM SR bin 6 bin 1
377 MC stat: Low-dM QCDCR, bin 13 bin 1
378 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, Top
379 MC stat: Low-dM QCDCR, bin 20 bin 1, Zto2Nu
380 MC stat: High-dM GCR, Nb3plus, recoil bin 4 bin 1
381 MC stat: Low-dM QCDCR, bin 33 bin 1, Top
382 MC stat: High-dM GCR, Nb3plus, recoil bin 1 bin 1
383 MC stat: Low-dM SR bin 30 bin 1, QCD
384 MC stat: Low-dM SR bin 10 bin 1
385 MC stat: Low-dM QCDCR, bin 22 bin 1, Top
386 MC stat: Low-dM SR bin 28 bin 1, Top
387 MC stat: Low-dM LLCR, bin 8 bin 1
388 MC stat: Low-dM GCR, bin 24 bin 1
389 MC stat: Low-dM GCR, bin 16 bin 1
390 MC stat: Low-dM LLCR, bin 16 bin 1
−2 −1 0 1 2
−0.1 0 0.1
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

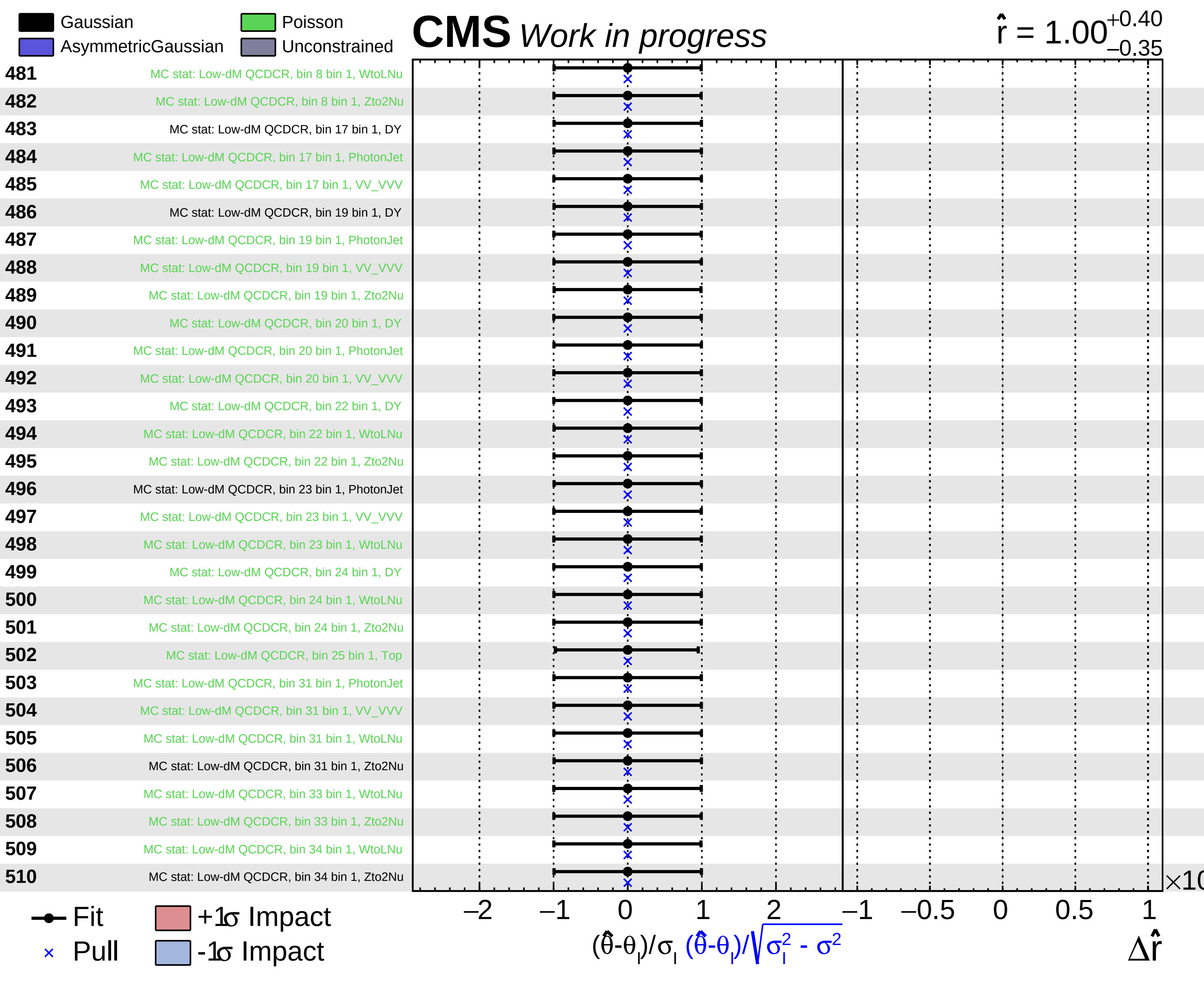


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.40}_{-0.35}$
391 MC stat: Low-dM QCDCR, bin 16 bin 1
392 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, WtoLNu
393 MC stat: Low-dM SR bin 6 bin 1
394 MC stat: Low-dM LLCR, bin 4 bin 1
395 MC stat: High-dM QCDCR, Nb1, recoil bin 5 bin 1
396 MC stat: Low-dM SR bin 12 bin 1
397 MC stat: High-dM SR bin 36 bin 1, Top_Nb2
398 MC stat: Low-dM GCR, bin 34 bin 1
399 MC stat: High-dM SR bin 54 bin 1, Zto2Nu_Nb3plus
400 MC stat: Low-dM SR bin 9 bin 1
401 MC stat: High-dM GCR, Nb3plus, recoil bin 3 bin 1
402 MC stat: Low-dM SR bin 30 bin 1, WtoLNu
403 MC stat: Low-dM SR bin 15 bin 1
404 MC stat: Low-dM LLCR, bin 12 bin 1
405 MC stat: Low-dM LLCR, bin 15 bin 1
406 MC stat: High-dM LLCR, Nb3plus, recoil bin 2 bin 1
407 MC stat: Low-dM GCR, bin 12 bin 1
408 MC stat: High-dM SR bin 47 bin 1, VV_VVV
409 MC stat: Low-dM QCDCR, bin 23 bin 1, Top
410 MC stat: High-dM SR bin 48 bin 1, Top_Nb3plus
411 MC stat: High-dM SR bin 36 bin 1, VV_VVV
412 MC stat: Low-dM QCDCR, bin 9 bin 1
413 MC stat: Low-dM QCDCR, bin 12 bin 1
414 MC stat: Low-dM SR bin 4 bin 1
415 MC stat: Low-dM SR bin 33 bin 1, Zto2Nu
416 MC stat: Low-dM QCDCR, bin 34 bin 1, Top
417 MC stat: Low-dM QCDCR, bin 15 bin 1
418 MC stat: Low-dM QCDCR, bin 23 bin 1, Zto2Nu
419 MC stat: Low-dM SR bin 30 bin 1, Zto2Nu
420 MC stat: High-dM SR bin 46 bin 1, Zto2Nu_Nb3plus
−2
−1
0
1
2
−0.05
0
0.05
×10
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
421 MC stat: Low-dM SR bin 13 bin 1
422 MC stat: Low-dM LLCR, bin 30 bin 1
423 MC stat: Low-dM GCR, bin 23 bin 1
424 MC stat: Low-dM GCR, bin 15 bin 1
425 MC stat: High-dM SR bin 43 bin 1, signal (1200, 500)
426 MC stat: Low-dM GCR, bin 32 bin 1, PhotonJet
427 MC stat: Low-dM LLCR, bin 20 bin 1
428 MC stat: Low-dM SR bin 33 bin 1, signal (1200, 500)
429 MC stat: High-dM SR bin 21 bin 1, VV_VVV
430 MC stat: Low-dM QCDCR, bin 29 bin 1
431 MC stat: Low-dM QCDCR, bin 26 bin 1
432 MC stat: Low-dM QCDCR, bin 18 bin 1
433 MC stat: High-dM QCDCR, Nb2, recoil bin 4 bin 1
434 MC stat: High-dM SR bin 44 bin 1, signal (1200, 500)
435 MC stat: Low-dM LLCR, bin 13 bin 1
436 MC stat: Low-dM GCR, bin 30 bin 1
437 MC stat: Low-dM SR bin 28 bin 1, VV_VVV
438 MC stat: Low-dM SR bin 28 bin 1, WtoLNu
439 MC stat: Low-dM SR bin 28 bin 1, Zto2Nu
440 MC stat: Low-dM SR bin 30 bin 1, Top
441 MC stat: High-dM SR bin 33 bin 1, VV_VVV
442 MC stat: Low-dM SR bin 25 bin 1, VV_VVV
443 MC stat: High-dM SR bin 44 bin 1, VV_VVV
444 MC stat: Low-dM QCDCR, bin 31 bin 1, Top
445 MC stat: Low-dM SR bin 33 bin 1, PhotonJet
446 MC stat: Low-dM LLCR, bin 28 bin 1
447 MC stat: High-dM SR bin 46 bin 1, VV_VVV
448 MC stat: Low-dM SR bin 30 bin 1, VV_VVV
449 MC stat: Low-dM SR bin 28 bin 1, DY
450 MC stat: Low-dM SR bin 28 bin 1, signal (1200, 500)
−2
−1
0
1
2
−0.02
0
0.02
×10
Fit
+1σ Impact
Pull
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
451 MC stat: Low-dM SR bin 28 bin 1, PhotonJet
452 MC stat: Low-dM SR bin 30 bin 1, PhotonJet
453 MC stat: Low-dM SR bin 30 bin 1, signal (1200, 500)
454 MC stat: Low-dM QCDCR, bin 6 bin 1
455 MC stat: Low-dM SR bin 30 bin 1, DY
456 MC stat: High-dM SR bin 43 bin 1, VV_VVV
457 MC stat: Low-dM QCDCR, bin 31 bin 1, DY
458 MC stat: Low-dM SR bin 33 bin 1, VV_VVV
459 QCD CR constraint, low-dM Nb1_b09
460 QCD CR constraint, low-dM Nb1_b10
461 QCD CR constraint, low-dM Nb1_b11
462 QCD CR constraint, low-dM Nb1_b14
463 QCD CR constraint, low-dM Nb1_b15
464 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, VV_VVV
465 MC stat: Low-dM GCR, bin 25 bin 1, Top
466 MC stat: Low-dM GCR, bin 32 bin 1, QCD
467 MC stat: Low-dM GCR, bin 32 bin 1, Top
468 MC stat: Low-dM GCR, bin 33 bin 1, PhotonJet
469 MC stat: Low-dM LLCR, bin 34 bin 1, DY
470 MC stat: Low-dM LLCR, bin 34 bin 1, QCD
471 MC stat: Low-dM LLCR, bin 34 bin 1, VV_VVV
472 MC stat: Low-dM LLCR, bin 34 bin 1, WtoLNu
473 MC stat: Low-dM LLCR, bin 34 bin 1, Zto2Nu
474 MC stat: Low-dM QCDCR, bin 4 bin 1, DY
475 MC stat: Low-dM QCDCR, bin 4 bin 1, PhotonJet
476 MC stat: Low-dM QCDCR, bin 4 bin 1, WtoLNu
477 MC stat: Low-dM QCDCR, bin 4 bin 1, Zto2Nu
478 MC stat: Low-dM QCDCR, bin 8 bin 1, PhotonJet
479 MC stat: Low-dM QCDCR, bin 8 bin 1, Top
480 MC stat: Low-dM QCDCR, bin 8 bin 1, VV_VVV
1.0 +0 −0
1.0 +0 −0
1.0 +0 −0
1.0 +0 −0
1.0 +0 −0
−2
−1
0
1
2
−1
0
1
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.40 −0.35
481 MC stat: Low-dM QCDCR, bin 8 bin 1, WtoLNu
482 MC stat: Low-dM QCDCR, bin 8 bin 1, Zto2Nu
483 MC stat: Low-dM QCDCR, bin 17 bin 1, DY
484 MC stat: Low-dM QCDCR, bin 17 bin 1, PhotonJet
485 MC stat: Low-dM QCDCR, bin 17 bin 1, VV_VVV
486 MC stat: Low-dM QCDCR, bin 19 bin 1, DY
487 MC stat: Low-dM QCDCR, bin 19 bin 1, PhotonJet
488 MC stat: Low-dM QCDCR, bin 19 bin 1, VV_VVV
489 MC stat: Low-dM QCDCR, bin 19 bin 1, Zto2Nu
490 MC stat: Low-dM QCDCR, bin 20 bin 1, DY
491 MC stat: Low-dM QCDCR, bin 20 bin 1, PhotonJet
492 MC stat: Low-dM QCDCR, bin 20 bin 1, VV_VVV
493 MC stat: Low-dM QCDCR, bin 22 bin 1, DY
494 MC stat: Low-dM QCDCR, bin 22 bin 1, WtoLNu
495 MC stat: Low-dM QCDCR, bin 22 bin 1, Zto2Nu
496 MC stat: Low-dM QCDCR, bin 23 bin 1, PhotonJet
497 MC stat: Low-dM QCDCR, bin 23 bin 1, VV_VVV
498 MC stat: Low-dM QCDCR, bin 23 bin 1, WtoLNu
499 MC stat: Low-dM QCDCR, bin 24 bin 1, DY
500 MC stat: Low-dM QCDCR, bin 24 bin 1, WtoLNu
501 MC stat: Low-dM QCDCR, bin 24 bin 1, Zto2Nu
502 MC stat: Low-dM QCDCR, bin 25 bin 1, Top
503 MC stat: Low-dM QCDCR, bin 31 bin 1, PhotonJet
504 MC stat: Low-dM QCDCR, bin 31 bin 1, VV_VVV
505 MC stat: Low-dM QCDCR, bin 31 bin 1, WtoLNu
506 MC stat: Low-dM QCDCR, bin 31 bin 1, Zto2Nu
507 MC stat: Low-dM QCDCR, bin 33 bin 1, WtoLNu
508 MC stat: Low-dM QCDCR, bin 33 bin 1, Zto2Nu
509 MC stat: Low-dM QCDCR, bin 34 bin 1, WtoLNu
510 MC stat: Low-dM QCDCR, bin 34 bin 1, Zto2Nu
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θI)/σI (θ̂-θI)/√(σI² - σ²)
Δr̂